ORIGINAL ARTICLE

Timing of PCI in Patients Undergoing Transcatheter Aortic-Valve Implantation

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ABSTRACT

BACKGROUND

Approximately one half of patients undergoing transcatheter aortic-valve implantation (TAVI) have concomitant coronary artery disease. Although percutaneous coronary intervention (PCI) is often performed before TAVI, the preferred treatment strategy has not been established.

METHODS

We conducted an international, open-label, randomized, noninferiority trial at 48 centers in Europe. Patients with severe aortic stenosis and coronary artery disease were randomly assigned in a 1:1 ratio to a strategy of either TAVI before PCI (TAVI-first group) or PCI before TAVI (PCI-first group). The primary end point was a composite of death from any cause; nonfatal myocardial infarction; ischemia-driven revascularization; rehospitalization related to the valve, procedure, or heart failure; or life-threatening, disabling, or major bleeding at 1 year after randomization. The noninferiority margin was 6.6 percentage points, with testing for noninferiority of TAVI first as compared with PCI first.

RESULTS

A total of 986 patients underwent randomization: 498 were assigned to the TAVI-first group and 488 to the PCI-first group. A primary end-point event occurred in 105 patients (22.2%) in the TAVI-first group and in 112 patients (24.2%) in the PCI-first group (risk difference, –2.0 percentage points; 95% confidence interval, –7.4 to 3.4; P<0.001 for noninferiority). Serious adverse events occurred in 264 patients in the TAVI-first group and in 273 patients in the PCI-first group.

CONCLUSIONS

Among patients with severe aortic stenosis and coronary artery disease, a strategy of TAVI before PCI was noninferior to a strategy of PCI before TAVI with respect to the primary end point at 1 year. (Funded by University Hospital Zurich and others; TAVI PCI ClinicalTrials.gov number, NCT04310046.)

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*A complete list of the TAVI PCI investigators is provided in the Supplementary Appendix, available at NEJM.org.

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Aortic stenosis is the most prevalent valvular heart disease and is associated with substantial morbidity and mortality.[1,2] Transcatheter aortic-valve implantation (TAVI) has become the standard treatment for patients with severe symptomatic aortic stenosis who are older than 65 to 70 years of age or are at prohibitive surgical risk.[1,2] Because aortic stenosis and coronary artery disease share several common risk factors, approximately one half of patients with severe aortic stenosis undergoing TAVI have obstructive coronary artery disease.[3-5] Observational studies have suggested an association between the presence and severity of coronary artery disease and adverse outcomes in patients who undergo TAVI.[6-10] Therefore, approximately 10 to 20% of patients undergoing TAVI also undergo percutaneous coronary intervention (PCI).[5,11,12] Current guidelines recommend angiography-guided PCI for patients with severe aortic stenosis who are undergoing TAVI and have concomitant coronary artery disease in proximal segments of main vessels.[1,2]

Performing PCI before TAVI represents the current standard treatment strategy, since it ensures coronary access before valve implantation and completes revascularization before the valve procedure. A TAVI-first strategy may offer advantages, including improved hemodynamic stability during subsequent PCI and avoidance of dual antiplatelet therapy at the time of TAVI; however, it also arouses concerns regarding subsequent coronary access. Given these competing considerations, the Optimal Timing of Transcatheter Aortic Valve Implantation and Percutaneous Coronary Intervention (TAVI PCI) trial was designed to assess whether a TAVI-first strategy would be noninferior to the current standard PCI-first strategy with respect to major clinical events.[13]

METHODS

TRIAL DESIGN AND OVERSIGHT

We conducted this investigator-initiated, international, open-label, randomized, noninferiority trial at 48 centers in Europe (Austria, France, Germany, Italy, the Netherlands, and Switzerland). The trial design has been described previously.[13] The TAVI PCI trial compared a strategy of TAVI followed by angiography-guided PCI within 1 to 45 days (TAVI-first group) with a strategy of angiography-guided PCI followed by TAVI within 1 to 45 days (PCI-first group) in patients with severe aortic stenosis and concomitant coronary artery disease.

The trial was performed in accordance with the principles of the Declaration of Helsinki. The trial protocol (available with the full text of this article at NEJM.org) was approved by the ethics committees at each participating center. The trial was coordinated and sponsored by the University Hospital Zurich and the University of Zurich. All the patients provided written informed consent. Details regarding the informed consent process are provided in the Supplementary Appendix (available at NEJM.org).

The members of a steering committee and an independent data and safety monitoring board oversaw the trial. An independent clinical-events committee whose members were unaware of the trial-group assignments adjudicated the key end points. The Institute for Pharmacology and Preventive Medicine (IPPMed, Cloppenburg, Germany) maintained the trial database, and the Stiftung Institut für Herzinfarktforschung (Ludwigshafen, Germany) performed the statistical analyses. Additional information regarding the trial oversight and author responsibilities is provided in the Supplementary Appendix. The trial was funded primarily through institutional research support from the Department of Cardiology, University Hospital Zurich, and the University of Zurich, which included funding for infrastructure and research personnel provided by the Andreas Grüntzig Clinical Research Center, Department of Cardiology, University Hospital Zurich. Other financial support was provided through an investigator-initiated research grant from Edwards Lifesciences. Edwards Lifesciences had no role in the design or conduct of the trial; the collection, management, analysis, or interpretation of the data; the writing, review, or approval of the manuscript; or the decision to submit the manuscript for publication. The authors vouch for the accuracy and completeness of the data and for the fidelity of the trial to the protocol. A detailed list of participating centers and investigators is provided in the Supplementary Appendix.

ELIGIBILITY

Adults 18 years of age or older were eligible to enroll in the trial if they had severe aortic steno-

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sis and concomitant coronary artery disease and if an interdisciplinary local heart team had established the indication for TAVI with a transcatheter heart valve (Edwards SAPIEN 3, SAPIEN 3 Ultra, or SAPIEN 3 Ultra RESILIA) and PCI. Severe symptomatic aortic stenosis was defined as an aortic-valve area of 1.0 cm^2 or less or a mean pressure gradient of 40 mm Hg or greater on echocardiography and at least one of the following criteria: dyspnea, angina symptoms, syncope, a left ventricular ejection fraction of less than 50%, symptoms (angina, dyspnea, or syncope) or a decrease in blood pressure on exercise testing, or the presence of high-risk criteria (i.e., peak transaortic velocity of >5.5 m per second, severe valve calcification, peak transaortic velocity progression of ≥0.3 m per second per year, or severe pulmonary hypertension with systolic pulmonary artery pressure of >60 mm Hg).[14] Additional details regarding the eligibility criteria are provided in the Supplementary Appendix.

Coronary artery disease was initially defined as at least one coronary-artery lesion with a diameter stenosis of 40 to 90% and an instantaneous wave-free ratio (or comparable resting indexes) of 0.89 or less or with a diameter stenosis of greater than 90% on the coronary angiogram (by visual estimation) in a coronary artery measuring at least 2.5 mm in diameter and Thrombolysis in Myocardial Infarction flow grade 3, deemed amenable to PCI within 45 days before or after TAVI. After the enrollment of 511 patients in October 2023, the steering committee approved a protocol amendment replacing routine physiology-guided PCI with an angiography-based strategy (with coronary artery disease defined as lesions with a diameter stenosis of ≥70% in a coronary artery measuring ≥2.5 mm in diameter); functional lesion assessment remained optional.[14] The decision of the steering committee regarding the protocol amendment was made without knowledge of treatment-specific outcome data and with the goal of increasing the rate of enrollment. The Synergy between Percutaneous Coronary Intervention with Taxus and Cardiac Surgery (SYNTAX) Score I was calculated for each patient. The SYNTAX Score I reflects a comprehensive angiographic assessment of the coronary vasculature; scores increase from 0 and have no defined upper limit, with higher scores indicating more-complex coronary artery anatomy.

RANDOMIZATION AND TREATMENT

Randomization was performed with the use of a secured, centralized Web-based system (S4Trials). Patients were randomly assigned in a 1:1 ratio to the TAVI-first group or the PCI-first group according to a computer-generated randomization schedule, with stratification according to trial center and the Society of Thoracic Surgeons Predicted Risk of Mortality risk category (low risk of death, <4%; intermediate risk of death, 4 to <10%; or high risk of death, ≥10%). Patients were considered to be enrolled in the trial after undergoing randomization. If a patient withdrew consent before completion of the 1-year follow-up period or was lost to follow-up, all data obtained up to the time of consent withdrawal or loss to follow-up were included in the analysis.

PCI was performed according to current guidelines with the use of standard interventional techniques.[15-18] Intravascular imaging, including intravascular ultrasonography and optical coherence tomography, was performed at the operator's discretion. All the patients received appropriate medical management and secondary prevention. A daily regimen of at least 80 mg of aspirin was prescribed indefinitely, except in patients receiving oral anticoagulation. After PCI, dual antiplatelet therapy was recommended for at least 3 months and preferably for 6 months, except in patients receiving oral anticoagulation, who received treatment with concomitant clopidogrel for at least 3 months.

TAVI was performed with standard techniques and in accordance with current guidelines.[1,2,19,20] The procedures used for the heart-valve prosthesis, associated delivery system, and all the components adhered to the respective instructions for use. After TAVI was performed, antiplatelet therapy with aspirin was administered unless patients were receiving oral anticoagulation, in which case they received oral anticoagulation only.

Follow-up was to be performed at hospital discharge (first and second hospitalization), at 3 months (with a window of ±14 days), and at 1 year (window, ±30 days). Although the 1-year follow-up was performed as an in-person visit, all other visits could occur in person or by telephone. All trial sites were monitored for adherence to the protocol and completeness of trial-related data capture.

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PRIMARY AND SECONDARY END POINTS

The primary end point was a composite of death from any cause; nonfatal myocardial infarction; ischemia-driven revascularization; rehospitalization related to the valve, procedure, or heart failure; or life-threatening, disabling, or major bleeding at 1 year after randomization. Prespecified secondary end points were assessed at hospital discharge (for the first and second hospitalizations), at 3 months, and at 1 year. Secondary end points included individual components of the primary end point and the following 11 individual end points: death from cardiovascular causes, stroke, target-lesion revascularization, target-vessel revascularization, stent thrombosis, major vascular complications, new-onset atrial fibrillation, new permanent pacemaker implantation, aortic-valve–related dysfunction leading to a repeat procedure, acute kidney injury, and bleeding (according to the Bleeding Academic Research Consortium [BARC] definition and the Valve Academic Research Consortium [VARC]–2 consensus document[21]). BARC types range from 0 to 5, with higher values indicating greater severity of bleeding. Secondary end points also included the following 4 composite end points: death from any cause or nonfatal myocardial infarction; death from any cause, nonfatal myocardial infarction, or ischemia-driven revascularization; death from any cause, nonfatal myocardial infarction, ischemia-driven revascularization, or rehospitalization related to the valve, procedure, or heart failure; and death from cardiovascular causes or nonfatal myocardial infarction.

Prespecified procedural end points were assessed at hospital discharge after the respective procedure. Procedural end points assessed after PCI included procedural success, amount of contrast medium, and use of catecholamines during PCI. Procedural end points assessed after TAVI included device success, use of catecholamines during TAVI, and conversion to open heart surgery. Quality of life (measured with the Kansas City Cardiomyopathy Questionnaire [KCCQ] and the Toronto Aortic Stenosis Quality of Life Questionnaire [TASQ]) and symptom status (according to Canadian Cardiovascular Society [CCS] and New York Heart Association [NYHA] classification) were assessed at 3 months and at 1 year. Detailed descriptions of the end-point definitions are provided in the Supplementary Appendix.

STATISTICAL ANALYSIS

On the basis of an estimated 15% incidence of a primary end-point event in each group at 1 year after randomization,[5,22-24] we determined that a sample of 934 patients (467 patients in each group) would provide the trial with 80% power to detect noninferiority of a TAVI-first strategy to a PCI-first strategy with respect to the primary end point, with a noninferiority margin of 6.6 percentage points.[5,25,26] To allow for dropout of 5% of the patients, a total of 986 patients were recruited. No interim analysis was performed.

Baseline and procedural characteristics were analyzed in the intention-to-treat population, to the extent that data were available. For analyses of clinical end points at 1 year, patients who did not have an end-point event within 365 days after randomization and did not complete the 1-year follow-up were excluded. The remaining patients constituted the modified intention-to-treat population which formed the basis for the statistical tests of the study hypotheses and the effect estimates presented. The primary end point was analyzed by calculating the percentage of patients with an occurrence of any of its components. Noninferiority of a TAVI-first strategy to a PCI-first strategy with respect to the primary end point was assessed with a Farrington–Manning test at a one-sided alpha level of 0.025. At the point at which noninferiority was shown, a prespecified superiority analysis was performed with the Farrington–Manning method. As supportive analyses, the hazard ratio and risk difference with corresponding 95% confidence intervals are reported with the use of 1-year event rates estimated with the product-limit method to account for patients with incomplete follow-up. Cumulative-incidence curves were used to visualize primary end-point events up to 1 year after randomization. In addition, supportive analyses were performed under different distributions of primary end-point events among patients with missing outcome data. The primary end point was also analyzed in two modified per-protocol populations. In the modified per-protocol population, patients were excluded if they did not meet the eligibility criteria, crossed over to the other treatment group, did not undergo TAVI, did not undergo PCI owing to logistic reasons or patient refusal, or did not undergo the procedures within the defined time window. In the second modified per-protocol population, patients

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were excluded if they did not meet the eligibility criteria or did not undergo PCI owing to administrative reasons. Only patients who had undergone randomization and had data available with regard to the primary end point at 1 year were included.

Effect sizes for secondary end points are presented as risk differences with corresponding 95% confidence intervals. Quality-of-life questionnaires (KCCQ and TASQ) were scored according to algorithms provided by the vendor. Adjustment for multiplicity was not performed, and the 95% confidence intervals should not be used to infer definitive treatment effects for secondary end points. No imputation of missing data was performed.

Results of prespecified subgroup analyses of the primary end point are presented as forest plots of the risk difference and 95% confidence intervals for comparisons between the two trial groups. Landmark analyses were performed for the primary end point, with assessment of events occurring before day 45 and those occurring on or after day 45 and assessment of events occurring through month 6 and those occurring after month 6; the cumulative incidence of events is shown graphically.

All analyses were performed with the use of SAS software, version 9.4 (SAS Institute). The statistical analysis plan is available with the trial protocol at NEJM.org.

RESULTS

PATIENTS

From September 2020 through August 2025, a total of 2705 patients with severe aortic stenosis and suspected coronary artery disease were screened. A total of 986 patients underwent randomization: 498 were assigned to the TAVI-first group and 488 to the PCI-first group (Fig. S1 in the Supplementary Appendix).

Follow-up at 1 year was completed in 473 patients (95.0%) in the TAVI-first group and in 463 patients (94.9%) in the PCI-first group. The modified per-protocol population consisted of 448 patients (90.0%) in the TAVI-first group and 447 patients (91.6%) in the PCI-first group (Fig. S2). The characteristics of the patients at baseline appeared to be well balanced between the two trial groups (Table 1 and Table S1). The representativeness of the trial population is shown in Table S2.

TREATMENT

Angiographic findings and procedural characteristics are summarized in Table 2 and Table S3. During the trial, 5 patients crossed over from the TAVI-first group to the PCI-first group, and 3 patients crossed over from the PCI-first group to the TAVI-first group. Reasons for crossover are provided in Table S4.

A total of 36 patients did not undergo TAVI, and 71 patients did not undergo PCI (Tables S5 and S6, respectively). Among these patients, in the TAVI-first group, PCI was not performed owing to a clinical decision of the operator in 13 patients and because of a negative functional lesion assessment in 15 patients. One patient had failed PCI, and 2 patients declined to undergo PCI. In the PCI-first group, PCI was not performed owing to a clinical decision of the operator in 5 patients and because of a negative functional lesion assessment in 3 patients. One patient had a failed PCI.

The types of transcatheter heart-valve prostheses used for implantation are listed in Table S7. The medical therapies are listed in Table S8.

The median time from randomization to the TAVI procedure was 8 days (interquartile range, 2 to 23) in the TAVI-first group, whereas the median time from randomization to the PCI procedure was 2 days (interquartile range, 1 to 12) in the PCI-first group. The median time between the two procedures was 27 days (interquartile range, 5 to 38) in the TAVI-first group and 13 days (interquartile range, 2 to 33) in the PCI-first group. TAVI and PCI procedures were performed during the same hospitalization in 370 of 968 patients with available data (38.2%). Procedural outcomes are shown in Table S9.

PRIMARY AND SECONDARY END POINTS

At 1 year after randomization, a primary endpoint event had occurred in 105 of 473 patients (22.2%) in the TAVI-first group and in 112 of 463 patients (24.2%) in the PCI-first group, for a risk difference of −2.0 percentage points (95% confidence interval [CI], −7.4 to 3.4; $P<0.001$ for noninferiority; $P=0.47$ for superiority) and a hazard ratio of 0.90 (95% CI, 0.69 to 1.18) (Table 3). Cumulative-incidence curves of the primary end

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Table 1. Characteristics of the Patients at Baseline.*

Characteristic	TAVI-First Group (N = 498)	PCI-First Group (N = 488)
Median age (IQR) — yr	82 (79–85)	82 (77–84)
Male sex — no. (%)	313 (62.9)	336 (68.9)
Median body-mass index (IQR)†	26.2 (23.7–29.7)	26.4 (24.1–29.4)
Median EuroSCORE II score (IQR) — %‡	3.1 (2.0–5.0)	3.2 (2.1–5.5)
Median STS-PROM score (IQR) — %§	3.1 (2.0–5.4)	3.3 (2.0–5.4)
New York Heart Association class — no. (%)		
I	42 (8.4)	41 (8.4)
II	205 (41.2)	171 (35.0)
III	236 (47.4)	245 (50.2)
IV	15 (3.0)	31 (6.4)
Canadian Cardiovascular Society class — no. (%)		
No angina	315 (63.3)	295 (60.5)
I	59 (11.8)	52 (10.7)
II	83 (16.7)	96 (19.7)
III	38 (7.6)	40 (8.2)
IV	3 (0.6)	5 (1.0)
Syncope — no. (%)	57 (11.4)	51 (10.5)
Medical history — no. (%)		
Hypertension	394 (79.1)	390 (79.9)
Diabetes	155 (31.1)	163 (33.4)
Previous PCI	98 (19.7)	118 (24.2)
Previous myocardial infarction	52 (10.4)	59 (12.1)
History of stroke or transient ischemic attack	51 (10.2)	58 (11.9)
Atrial fibrillation	79 (15.9)	95 (19.5)
Peripheral artery disease	72 (14.5)	61 (12.5)
Pulmonary hypertension	30 (6.0)	38 (7.8)
Chronic obstructive pulmonary disease	36 (7.2)	39 (8.0)
Previous malignant disease	110 (22.1)	89 (18.2)
Permanent pacemaker	32 (6.4)	41 (8.4)

* Percentages may not total 100% because of rounding. IQR denotes interquartile range, PCI percutaneous coronary intervention, and TAVI transcatheter aortic-valve implantation.
† The body-mass index is the weight in kilograms divided by the square of the height in meters.
‡ The European System for Cardiac Operative Risk Evaluation (EuroSCORE) II score ranges from 0 to 100%, with higher scores indicating a higher risk of death.
§ The Society of Thoracic Surgeons–Predicted Risk of Mortality (STS-PROM) score ranges from 0 to 100%, with higher scores indicating a greater risk of death within 30 days after the procedure.

point through 1 year after randomization are shown in Figure 1.

In a conservative sensitivity analysis, 25 primary end-point events were allocated among the 50 patients with missing end-point data, with 20 events assigned to the TAVI-first group and 5 to the PCI-first group. Statistical evidence for noninferiority was preserved. The estimated risk difference that was based on the product-limit estimator for the cumulative incidence of events at 1 year was –1.7 percentage points (95% CI, –7.0 to 3.6). This finding was consistent with that in the primary analysis. Analysis of the primary end point in the modified per-protocol

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Table 2. Angiographic Findings and Procedural Characteristics.

Variable	TAVI-First Group (N = 498)	PCI-First Group (N = 488)
Angiographic findings		
SYNTAX Score I*		
Median score (IQR)	10 (6–16)	11 (7–17)
No. of patients with data	495	487
No. of vessels treated with PCI — no. (%)		
0	63 (12.7)	17 (3.5)
1	321 (64.5)	327 (67.0)
2	100 (20.1)	117 (24.0)
3	14 (2.8)	27 (5.5)
Location of lesions treated with PCI — no. (%)		
Left main coronary artery	20 (4.0)	32 (6.6)
Left anterior descending coronary artery	300 (60.2)	318 (65.2)
Left circumflex coronary artery	103 (20.7)	142 (29.1)
Right coronary artery	135 (27.1)	150 (30.7)
Chronic total occlusion — no./total no. (%)	36/497 (7.2)	23/488 (4.7)
No. of patients undergoing functional lesion assessment — no. (%)	168 (33.7)	158 (32.4)
PCI procedure		
Median no. of days from randomization to PCI (IQR)	39 (11–58)	2 (1–12)
Radial access — no./total no. (%)	302/441 (68.5)	336/474 (70.9)
Median no. of stents used per patient (IQR)	2 (1–2)	2 (1–2)
Median total stent length (IQR) — mm	33 (19–58)	35 (18–57)
Intravascular ultrasonography — no./total no. (%)	92/441 (20.9)	114/474 (24.1)
Optical coherence tomography — no./total no. (%)	19/441 (4.3)	15/474 (3.2)
Second PCI performed — no./total no. (%)	27/441 (6.1)	17/474 (3.6)
TAVI procedure		
Median no. of days from randomization to TAVI (IQR)	8 (2–23)	25 (5–42)
Balloon-expandable transcatheter heart valve — no./total no. (%)	467/486 (96.1)	440/464 (94.8)

* The Synergy between Percutaneous Coronary Intervention with Taxus and Cardiac Surgery (SYNTAX) Score I reflects a comprehensive angiographic assessment of the coronary vasculature; scores increase from 0 and have no defined upper limit, with higher scores indicating more complex coronary-artery anatomy.

population was also consistent with the modified intention-to-treat population. Supportive analyses are shown in Tables S10 and S11. Results of subgroup analyses of the primary end point are shown in Figure 2.

Secondary end points are shown in Table 3. Death occurred in 45 of 472 patients (9.5%) in the TAVI-first group and in 44 of 462 patients (9.5%) in the PCI-first group (risk difference, 0.0 percentage points; 95% CI, −3.8 to 3.8). The causes of death are listed in Table S12. Nonfatal myocardial infarction occurred in 14 of 431 patients (3.2%) in the TAVI-first group and in 16 of 423 patients (3.8%) in the PCI-first group (risk difference, −0.5 percentage points; 95% CI, −3.0 to 1.9) (Table S13). Ischemia-driven revascularization occurred in 6 of 428 patients (1.4%) in the TAVI-first group and in 9 of 419 patients (2.1%) in the PCI-first group (risk difference, −0.7 percentage points; 95% CI, −2.5 to 1.0). Rehospitalization related to the valve, procedure, or heart failure occurred in 39 of 440 patients (8.9%) in the TAVI-first group and in 30 of 425 patients (7.1%) in the PCI-first group (risk difference,

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Table 3. Primary and Secondary End Points.*

End Point	TAVI-First Group (N=498)	PCI-First Group (N=488)	Risk Difference (95% CI)
	no./total no. (%)		*percentage points*
Primary end point at 1 yr†			
Death from any cause; nonfatal myocardial infarction; ischemia-driven revascularization; rehospitalization related to the valve, procedure, or heart failure; or life-threatening, disabling, or major bleeding	105/473 (22.2)	112/463 (24.2)	−2.0 (−7.4 to 3.4)
Secondary end points at 1 yr‡			
Death from any cause	45/472 (9.5)	44/462 (9.5)	0.0 (−3.8 to 3.8)
Nonfatal myocardial infarction	14/431 (3.2)	16/423 (3.8)	−0.5 (−3.0 to 1.9)
Ischemia-driven revascularization	6/428 (1.4)	9/419 (2.1)	−0.7 (−2.5 to 1.0)
Rehospitalization related to the valve, procedure, or heart failure	39/440 (8.9)	30/425 (7.1)	1.8 (−1.8 to 5.4)
Life-threatening, disabling, or major bleeding	29/438 (6.6)	42/431 (9.7)	−3.1 (−6.8 to 0.5)
Death from any cause or nonfatal myocardial infarction	56/472 (11.9)	56/462 (12.1)	−0.3 (−4.4 to 3.9)
Death from any cause, nonfatal myocardial infarction, or ischemia-driven revascularization	60/472 (12.7)	63/462 (13.6)	−0.9 (−5.3 to 3.4)
Death from any cause, nonfatal myocardial infarction, ischemia-driven revascularization, or rehospitalization related to the valve, procedure, or heart failure	86/473 (18.2)	83/462 (18.0)	0.2 (−4.7 to 5.1)
Death from cardiovascular causes	28/455 (6.2)	29/450 (6.4)	−0.3 (−3.5 to 2.9)
Death from cardiovascular causes or nonfatal myocardial infarction	40/456 (8.8)	42/449 (9.4)	−0.6 (−4.3 to 3.2)
Stroke	22/435 (5.1)	24/424 (5.7)	−0.6 (−3.6 to 2.4)
Target-vessel revascularization	6/428 (1.4)	8/419 (1.9)	−0.5 (−2.2 to 1.2)
Target-lesion revascularization	5/428 (1.2)	6/419 (1.4)	−0.3 (−1.8 to 1.3)
Stent thrombosis	1/428 (0.2)	1/420 (0.2)	0.0 (−0.7 to 0.6)
Major vascular complications	8/432 (1.9)	13/422 (3.1)	−1.2 (−3.3 to 0.9)
Any bleeding			
BARC type 2	27/434 (6.2)	28/422 (6.6)	−0.4 (−3.7 to 2.9)
BARC type 3a	15/432 (3.5)	29/426 (6.8)	−3.3 (−6.3 to −0.4)
BARC type 3b	10/431 (2.3)	12/421 (2.9)	−0.5 (−2.7 to 1.6)
BARC type 3c	2/429 (0.5)	1/420 (0.2)	0.2 (−0.6 to 1.0)
BARC type 5	3/431 (0.7)	2/421 (0.5)	0.2 (−0.8 to 1.2)
New-onset atrial fibrillation	38/433 (8.8)	25/422 (5.9)	2.9 (−0.6 to 6.3)
New permanent pacemaker implantation	69/439 (15.7)	59/428 (13.8)	1.9 (−2.8 to 6.7)
Aortic-valve dysfunction requiring repeat procedure	0/428 (0)	0/419 (0)	—
Acute kidney failure	20/430 (4.7)	27/425 (6.4)	−1.7 (−4.8 to 1.4)

* The treatment effect is the risk difference with 95% confidence interval for an end-point event in the TAVI-first group as compared with the PCI-first group. Bleeding was defined according to the Bleeding Academic Research Consortium (BARC) definition and the Valve Academic Research Consortium (VARC)–2 consensus document; BARC types range from 0 to 5, with higher values indicating greater severity of bleeding.

† The primary analysis was performed according to a modified intention-to-treat principle that included all the patients who had undergone randomization and had data available with regard to the primary end point at 1 year. $P<0.001$ for noninferiority, and $P=0.47$ for superiority.

‡ For secondary end points, the widths of the confidence intervals have not been adjusted for multiplicity and should not be used to infer definitive treatment effects.

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1.8 percentage points; 95% CI, −1.8 to 5.4). The incidence of life-threatening, disabling, or major bleeding was 6.6% in the TAVI-first group and 9.7% in the PCI-first group (risk difference, −3.1 percentage points; 95% CI, −6.8 to 0.5). Details regarding symptom status (according to NYHA and CCS classification) and quality-of-life outcomes are provided in Table S14. Post hoc landmark analyses are shown in Figure S3.

SAFETY

A serious adverse event occurred in 264 of 498 patients (53.0%) in the TAVI-first group and in 273 of 488 patients (55.9%) in the PCI-first group. The risk difference was −2.9 percentage points (95% CI, −9.1 to 3.3) (Table S15).

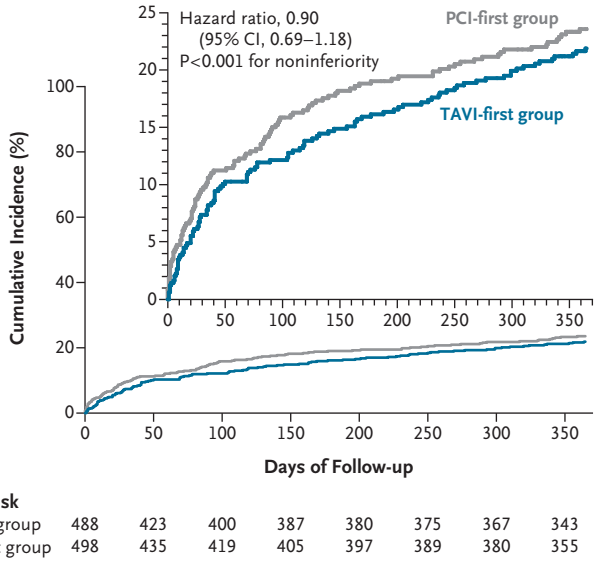


Figure 1. Cumulative Incidence of a Primary End-Point Event.

The primary end point was a composite of death from any cause; nonfatal myocardial infarction; ischemia-driven revascularization; rehospitalization related to the valve, procedure, or heart failure; or life-threatening, disabling, or major bleeding at 1 year after randomization. Bleeding was defined according to the Bleeding Academic Research Consortium (BARC) definition and the Valve Academic Research Consortium (VARC)–2 consensus document.[21] The inset shows the same data on an enlarged y axis. PCI denotes percutaneous coronary intervention, and TAVI transcatheter aortic-valve implantation.

DISCUSSION

In the TAVI PCI trial involving patients with severe aortic stenosis and concomitant coronary artery disease, a TAVI-first strategy was noninferior to a PCI-first strategy with respect to the primary composite end point — death from any cause; nonfatal myocardial infarction; ischemia-driven revascularization; rehospitalization related to the valve, procedure, or heart failure; or life-threatening, disabling, or major bleeding at 1 year after randomization. The overall occurrence of serious adverse events appeared to be similar in the two groups.

Coronary artery disease is common among patients with severe aortic stenosis and has been associated with adverse clinical outcomes.[6,7,27] In patients undergoing surgical aortic-valve replacement, coronary-artery bypass grafting is recommended for clinically significant coronary artery disease.[1,28,29] Current guidelines also recommend coronary revascularization in patients undergoing TAVI if they have severe aortic stenosis and concomitant coronary artery disease.[1,2] However, the benefit of PCI in this context remains uncertain, and randomized trials have yielded inconsistent results. In the NOTION-3 (Nordic Aortic Valve Intervention) trial, PCI performed before, during, or after TAVI was associated with a lower risk of a composite of death from any cause, myocardial infarction, or urgent revascularization than a conservative strategy.[8] In the PRO-TAVI (Percutaneous Coronary Intervention before Transcatheter Aortic Valve Implantation) trial, deferral of PCI was noninferior to the use of PCI before TAVI with respect to the composite of death from any cause, myocardial infarction, stroke, and major bleeding at 1 year, findings that suggest that PCI may be safely deferred in selected patients.[26] The ACTIVATION (Percutaneous Coronary Intervention Prior to Transcatheter Aortic-Valve Implantation) trial was terminated early and therefore did not provide conclusive evidence regarding the benefit of performing PCI before TAVI. Collectively, these trials addressed whether PCI should be performed or deferred, but they did not directly compare the sequence of TAVI and PCI when both procedures were planned.[30]

Treatment strategies have become increasingly relevant since TAVI is being used more often in a low-risk and younger patient population. Although the performance of PCI before TAVI remains the predominant strategy in many centers because it facilitates coronary-artery access and reduces the ischemic burden associated with rapid ventricular pacing and transcatheter

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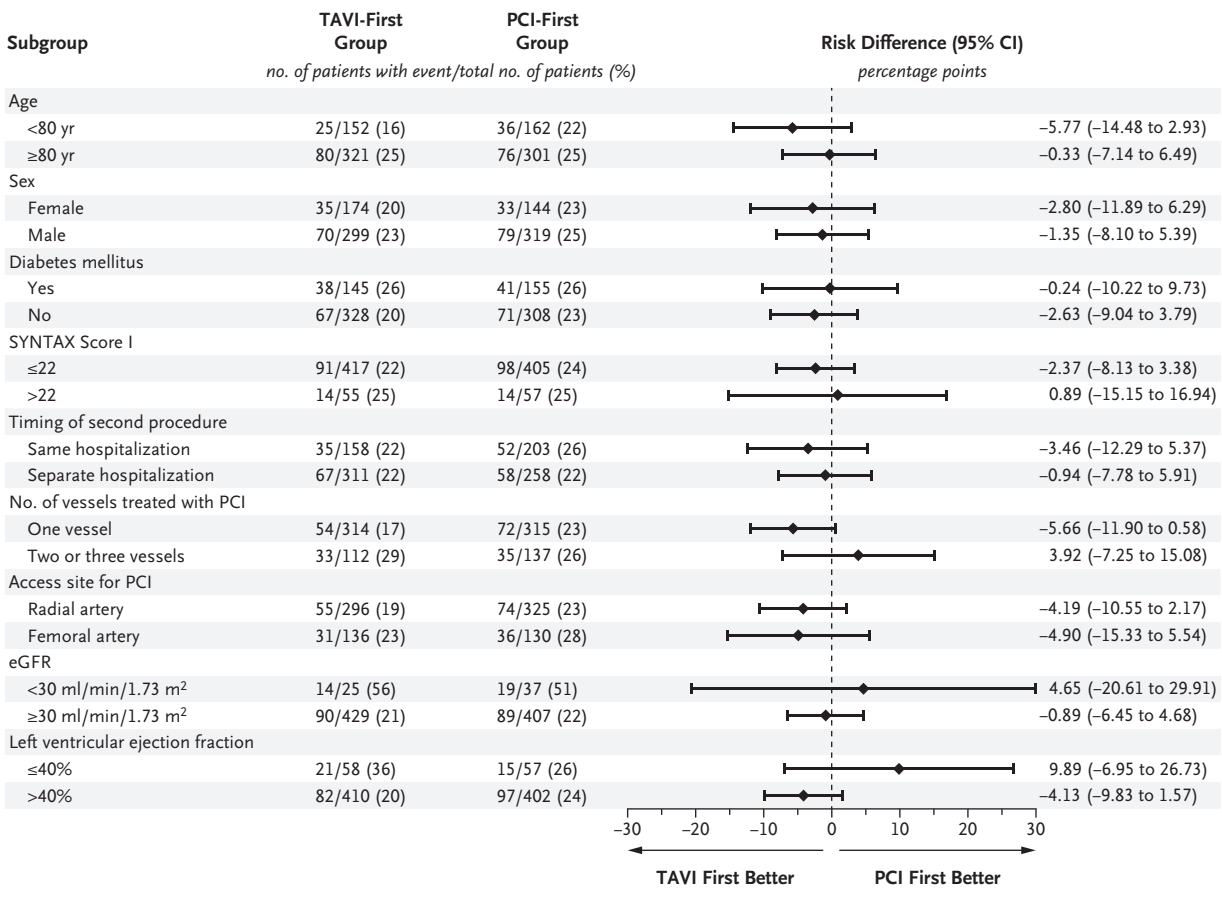


Figure 2. Subgroup Analyses of the Primary End Point.

The primary end point was a composite of death from any cause; nonfatal myocardial infarction; ischemia-driven revascularization; rehospitalization related to the valve, procedure, or heart failure; or life-threatening, disabling, or major bleeding at 1 year after randomization. The analyses are based on the modified intention-to-treat population. The widths of the confidence intervals have not been adjusted for multiplicity and should not be used to infer definitive treatment effects. The Synergy between Percutaneous Coronary Intervention with Taxus and Cardiac Surgery (SYNTAX) Score I reflects a comprehensive angiographic assessment of the coronary vasculature; scores increase from 0 and have no defined upper limit, with higher scores indicating more-complex coronary-artery anatomy. The term eGFR denotes estimated glomerular filtration rate.

heart-valve deployment, previous randomized trials did not include a TAVI-first strategy as a comparator. In the NOTION-3 trial, PCI was performed as a staged procedure before TAVI in most patients; however, concomitant PCI or PCI up to 2 days after TAVI was also permitted.[8] In the PRO-TAVI and ACTIVATION trials, PCI before TAVI was compared with an initial conservative strategy.[26,30] Observational studies have suggested that PCI after TAVI is feasible and safe, particularly after implantation of balloon-expandable transcatheter heart valves.[31,32] Furthermore, the risk of bleeding may be lower when dual antiplatelet therapy is initiated after TAVI than when it is initiated before TAVI.[26] The TAVI PCI trial provided evidence that a TAVI-first strategy was noninferior to a PCI-first strategy among patients with severe aortic stenosis and concomitant coronary artery disease who were selected to undergo both procedures. The higher-than-expected incidence of end-point events that was observed in both groups may reflect the fact that our trial population was older and had a greater burden of coexisting conditions than we had originally anticipated.

The number of PCI procedures performed was lower in the TAVI-first group than in the PCI-first group. This finding suggests that per-

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forming TAVI first may have altered subsequent clinical decision making in these patients. The TAVI PCI trial randomized treatment strategies rather than procedural sequences alone. Furthermore, since major concerns regarding coronary-artery access after TAVI have been raised, the fact that PCI was largely successful in patients undergoing TAVI with a balloon-expandable valve is worth noting, and PCI failure owing to the transcatheter heart-valve prosthesis seemed to be uncommon.

Although the incidence of life-threatening, disabling, or major bleeding appeared to be similar in the two groups, there may have been a difference in BARC type 3a bleeding events. BARC type 3a bleeding appeared to be more frequent in the PCI-first group than in the TAVI-first group. This finding may in part reflect the higher exposure to dual antiplatelet therapy at the time of TAVI, and a shorter duration of dual antiplatelet therapy may reduce the risk of bleeding. Whether the use of an abbreviated dual antiplatelet therapy could mitigate bleeding after a PCI-first strategy while maintaining ischemic protection warrants evaluation in future prospective studies.

Determining the clinical relevance of coronary artery disease remains challenging in patients with severe aortic stenosis because noninvasive and invasive indexes of ischemia have not been validated in this population, and functional coronary lesion assessment may be limited by reduced hyperemic coronary flow.[33] Current guidelines primarily recommend angiography-based assessment of the severity of coronary artery disease in patients with severe aortic stenosis.[1,2] In keeping with these recommendations, randomized trials, including NOTION-3, PRO-TAVI, and ACTIVATION, relied predominantly on coronary angiography to guide PCI.[8,26,30] Likewise, in the TAVI PCI trial, coronary artery disease was assessed predominantly by coronary angiography, with intracoronary physiological assessment and imaging performed at the operator's discretion.

Some limitations of the trial need to be considered. Women were underrepresented, which may limit generalizability of the findings. Patients with acute myocardial infarction, previous coronary-artery bypass grafting, or previously treated aortic stenosis were excluded. Accordingly, the findings should not be extrapolated to these populations. Patients with complex coronary artery disease (defined as a SYNTAX Score I of ≥33) were excluded, and overall coronary anatomical complexity was low (median SYNTAX Score I, 10), which may limit the applicability of the findings to patients with more complex coronary anatomies. PCI was guided predominantly by coronary angiography with intracoronary physiological assessment and imaging used at the operator's discretion. The higher-than-anticipated numbers of end-point events reduced the statistical power to detect noninferiority for the prespecified margin and therefore represent a limitation of the trial. Although the upper boundary of the observed confidence interval of the risk difference (3.4 percentage points) remains below the noninferiority margin (6.6 percentage points), it still permits a relative excess of events with a TAVI-first strategy. The findings may not be generalizable to patients undergoing nontransfemoral TAVI or to those who receive supra-annular prostheses.

Among patients with severe aortic stenosis and concomitant coronary artery disease who were selected to undergo TAVI and PCI, a TAVI-first strategy was noninferior to a PCI-first strategy with respect to the primary end point at 1 year.

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